

Stakeholder Engagement Plan — Bridging the Tokenization Gap in African LLMs via SuperBPE

Challenge 4 submission · Team Manda · TRI AI Cohort 10 · Project 3

Field	Entry
Team	Law Adagbon — solo participant (no additional team members, no assigned mentor)
Related documents	Challenge 1 problem statement; Challenge 2 Data Card (MATC v0.1); Challenge 3 Impact Statement
Engagement methods	World Café and Delphi, both selected from the Engage2020 Action Catalogue

1. Stakeholder map, and why these two groups

Our map identifies six groups with a stake in this project: speakers and language custodians of the languages we cover; developers and researchers building African-language systems; end users of downstream products; publishers and authors whose text sits in our source corpora; language academies and education authorities; and the cohort's own mentors and evaluators.

We have selected the first two for detailed engagement, on two grounds. They sit at opposite ends of the artefact — one group's language is being encoded, the other group will decide whether the encoding is used at all — and each holds knowledge we cannot obtain by any other means. No amount of measurement tells us whether a learned superword token is a real expression in Igbo; only an Igbo speaker can say. No corpus statistic tells us which languages a working developer actually needs covered.

An affordance worth stating up front

Most AI artefacts cannot be shown to the people they affect. A model's weights are unreadable. Ours is different: the deliverable is a list of at most 20,000 short, human-readable strings. A speaker can look at what the tokenizer decided counts as a unit of their language and say plainly that it is wrong. Both engagements below are built around that fact, which is what lets us consult on evidence rather than on intentions.

2. Stakeholder Group A — speakers and language custodians

Item	Detail
Who	First-language speakers of the languages in our corpus, weighted toward people who work with the written form: language teachers, translators, writers and editors, community radio staff, and members of language academies or orthography committees where they exist.
Their stake	Their language is being encoded into infrastructure that will outlive our involvement. Decisions we make about diacritics, dialect and register become defaults for anything built on the vocabulary.
Their likely concerns	That tone marks and underdots are treated as optional decoration; that a standard or urban variety is presented as the whole language; that their writing was taken from the web without consent; that consultation is cosmetic and the decisions are already made.

Method: World Café

Why this method. The Action Catalogue lists the World Café under dialogue and participatory research, for settings with many participants, no requirement of consensus, and a need for lateral discussion rather than question-and-answer. It suits us for three specific reasons. It

handles several languages in one session, because each table can work in one language with a speaker as host. Rotation exposes participants to each other's reasoning, which surfaces disagreements between varieties that a one-to-one interview would never reveal. And unlike a survey, it gives us reasons rather than ratings — we need to know *why* a token is wrong, not merely that it scored badly.

Why not the alternatives. A focus group would seat one language at one table and lose the cross-variety comparison. A survey would let us reach more people but would return judgements without justifications, which is the part we cannot reconstruct ourselves. A citizens' jury aims at a verdict; we are not asking anyone to rule on a proposition.

How it runs

— **Format.** One session of roughly two hours, run in person where a language community is concentrated and online where speakers are dispersed. Tables of four to six, one language per table, hosted by a speaker of that language. Three rounds of about twenty-five minutes, participants rotating between rounds, host remaining.

— **Language of the session.** Each table is conducted in the language under discussion, not in English. A session about whether our tokenizer respects Yorùbá, held entirely in English, would answer its own question.

— **Materials.** Printed extracts from the learned vocabulary: the longest superword tokens, the tokens that occur most often, and a set sampled at random. Alongside them, worked examples of how a sentence is segmented, before and after.

— **Harvest.** Table notes captured on the sheets themselves, then dot voting in plenary to rank the issues raised. Nothing recorded is discarded, whether or not it ranks highly.

What we will ask

- Which of these tokens are real units of the language, and which are fragments that mean nothing?
- Where a token fuses several words, is that a genuine fixed expression, or an accident of our corpus?
- Which written registers are missing from what you see here — and what do people actually write in, day to day?
- Where would losing tone marks or underdots change the meaning of these specific examples?
- What would make you unwilling to have your community's text in a corpus like this?

How the feedback changes the dataset and the model

Finding	Change it triggers
Tokens judged meaningless fragments in one language	Re-examine that language's sampling weight and source mix; a high rate of nonsense tokens usually means too little or too narrow text.
Superwords judged accidental rather than idiomatic	Lower the transition point between subword and superword merging, or cap token length, so fewer cross-word merges are learned.
Registers reported missing	Adjust the register quotas in the Data Card and seek permissioned sources for the gap.
Diacritic examples where meaning changes	Added to the held-out evaluation set, so the cost of ASCII flattening is measured on cases speakers identified rather than cases we guessed.
Objections to a source	Trigger the removal route committed to in the Data Card: drop the source, retrain, reissue.

3. Stakeholder Group B — African-language AI developers and researchers

Item	Detail
Who	Engineers and researchers building African-language systems: local startups, university NLP groups, and participants in open research communities working on the continent's languages.
Their stake	They are the adopters. A tokenizer that nobody integrates changes nothing, and they carry the consequences of any flaw we ship.
Their likely concerns	Whether the vocabulary covers the languages their users speak; whether gains hold on their own data rather than a leaderboard; licensing clarity sufficient for commercial use; the cost of switching tokenizers at all.

Method: Delphi

Why this method. This group is expert, small, opinionated and scattered across many countries and time zones. The Action Catalogue lists Delphi for exactly this situation: structured, iterative, remote consultation with controlled feedback between rounds, converging without requiring anyone in a room together. Two features matter for us particularly. Anonymity between rounds means a junior engineer's assessment carries the same weight as a well-known researcher's, which a live call does not deliver. And iteration lets us put our own trade-offs to the panel and see whether initial disagreement survives seeing others' reasoning.

Why not the alternatives. A workshop would favour whoever could attend and whoever speaks most. Interviews would give us a series of opinions with no way to tell a widely held view from a strongly held one. A hackathon would produce code, not the prioritisation judgement we need.

How it runs

- **Panel.** Twelve to twenty participants, recruited to span regions, language families and organisation types, with deliberate inclusion of people building for languages our corpus covers thinly.
- **Round 1 — open.** Short written questionnaire: which languages and registers matter most in your work, what would make you adopt or reject a new tokenizer, what would you need to see measured?
- **Round 2 — ranked.** We return the anonymised, grouped responses with our own constraints stated plainly, including the 20,000-token ceiling and the trade-off that vocabulary spent on one language is not available to another. Participants rank priorities and comment on the trade-offs.
- **Round 3 — convergence.** We circulate the distribution of rankings and ask participants to revise or hold their position and say why. Persistent disagreement is reported as disagreement rather than averaged away.

How the feedback changes the dataset and the model

Finding	Change it triggers
Ranked language priorities	Direct input to sampling weights and to which languages we commit to covering rather than merely touching.
Required evaluation measures	Added to our reporting: if adopters need per-language fertility on their own domain, we publish that rather than a single average.
Licensing thresholds for adoption	Determines which sources we can include at all, since a corpus component that blocks commercial use blocks the adopters we most want.

Finding	Change it triggers
Integration friction	Shapes the released interface and documentation, so that adoption cost is not the reason the work goes unused.

4. Conducting engagement responsibly

Consultation can itself be extractive. These commitments are part of the plan, not an afterthought to it:

- **Participants are compensated** for their time at a fair local rate. Expert judgement about a language is expert labour, and treating it as a free contribution because the participants are community members rather than professionals would reproduce the extraction this project exists to argue against.
- **We consult before decisions are locked.** Engagement scheduled after the corpus is frozen would be ratification, not consultation. Group A's session is placed before the final corpus freeze and Group B's first round before sampling weights are fixed.
- **We report back.** Every participant receives a short summary of what was heard, what we changed as a result, and what we did not change and why. The last of those is the part that makes the exercise honest.
- **Findings are recorded whether or not we act on them.** Constraints of time and vocabulary size mean we will not be able to act on everything; unactioned findings are published in the Data Card as known gaps rather than dropped.
- **Consent and data handling.** Participation is voluntary and withdrawable; notes are held without personal identifiers; nothing said in a session is attributed to a named individual without permission.

5. Sequencing

Phase	Activity	Feeds into
Before sampling weights are fixed	Delphi Round 1 issued to the developer panel	Language and register priorities
Draft vocabulary available	World Café with speakers, using printed vocabulary extracts	Token quality, register gaps, diacritic test cases
Between rounds	Delphi Rounds 2 and 3 with our constraints disclosed	Sampling weights, evaluation measures, licensing decisions
Before release	Findings written into the Data Card; summaries returned to all participants	Final corpus, final vocabulary, published limitations

6. How we will know the engagement worked

Not by attendance. We will judge it by whether it changed anything: the number of concrete design decisions altered as a direct result, recorded with the finding that prompted each; whether participants who reviewed the vocabulary would use it for their own language; and whether any group told us something we had not already written down. An engagement that confirms everything we already believed is more likely to have been badly designed than to have been reassuring.